



DesignWare® Cores

SuperSpeed USB 3.0 Controller

Installation Guide

*DWC_usb3 – **Product Codes***

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Revision History

The following table provides the history of changes to this document.

Date	Version	Description
May 2018	3.30b	No changes to this document
February 2018	3.30a	Updated: ■ “Downloading DWC_usb3” on page 13 ■ Tool versions in Table 1-4 on page 16 ■ “PHY GTECH and Behavioral Models” on page 15 ■ “Download and Install AMBA SVT Verification IP Components” on page 18 Removed references to the following: ■ VERA (not needed for the new PVE) ■ MTI and NCV (no longer supported)
February 2017	3.20a	Added: ■ Note on 64-bit binary requirement for all tools in “Verifying Supported Verification IP Components and Tool Versions” on page 16 ■ Licensing and product names for Automotive Package offering Updated: ■ “DWC USB 3.0 Controller Licenses” on page 11 ■ Tool versions in Table 1-4 on page 16 (“Supported Tool Versions”) ■ “Downloading and Installing coreConsultant” on page 19 Removed: ■ References to encrypted source files; all customers are required to have a valid source license set up before installing the .run file
March 2016	3.10a	Added: ■ “Download and Install AMBA SVT Verification IP Components” on page 18 Updated: ■ Tool versions in Table 1-4 on page 16 (“Supported Tool Versions”) ■ “Download and Install USB 3.0 Verification IP Components” on page 18 ■ “Downloading and Installing coreConsultant” on page 19
May 2015	3.00a	Updated tool versions in Table 1-4 on page 16 (“Supported Tool Versions”) and “Downloading and Installing Verification IP Components” on page 17

Date	Version	Description
September 2014	2.90a	Updated tool versions in Table 1-4 on page 16 (“Supported Tool Versions”) and “Downloading and Installing Verification IP Components” on page 17
June 2014	2.80a	Updated the download instructions for USB 3.0 Verification IP in “Downloading and Installing Verification IP Components” on page 17
February 2014	2.80a	Updated tool versions in Table 1-4 on page 16 (“Supported Tool Versions”), and “Downloading and Installing Verification IP Components” on page 17
December 2013	2.70	Added details on MPHY model in “PHY GTECH and Behavioral Models” on page 15 The USB 3.0 VIP (usb_svt) is no longer included in the DWC_usb3 image. It must be downloaded and installed separately. (See “Downloading and Installing Verification IP Components” on page 17.) The AMBA VIP is no longer included in the DWC_usb3 image. It must be downloaded and installed separately. (See “Downloading and Installing Verification IP Components” on page 17.) Updated Table 1-4, “Supported Tool Versions”
July 2013	2.60a	Added “Product Codes” on page 8 Removed Linux 4.0/32 bit from Table 1-2, “System Requirements” Updated Table 1-4, “Supported Tool Versions”
November 2012	2.50a	Updated Table 1-4 , “Supported Tool Versions”
August 2012	2.40a	Updated Table 1-4 , “Supported Tool Versions”
July 2012	2.30a	No updates for this release
April 2012	2.20a	Added: - PHY versions tested (see “PHY GTECH and Behavioral Models” on page 15) Updated: - Title page (list of base products and add-ons) Table 1-3, “Licenses for DWC USB 3.0 Controller” Table 1-4, “Supported Tool Versions” Table 1-5, “Environment Variables for DWC USB 3.0 Controller” Removed “Environment Variables” appendix (duplicate information from Table 1-5)
January 2012	2.10a	- Changed the name of the PDF filename for this Installation Guide from “usb3_install.pdf” to “DWC_usb3_install.pdf” - Table “Supported Tool Versions”: Updated
October 2011	2.00a	Table “Supported Tool Versions”: Updated
August 2011	1.92a	- Table “Supported Tool Versions”: Updated - Table “Licenses for DWC USB 3.0 Controller”: Updated
August 2011	1.90a	Table “Supported Tool Versions”: Updated

Date	Version	Description
January 2011	1.80a	■ Restructured the Installation Guide (split out sections into separate chapters; created separate environment setup section for each topic) ■ Added the following Appendices: - “Troubleshooting and Support” on page 21 - “Environment Variables” on page 23 (no longer an appendix) - “Example Setup File” on page 27 - “Options for .run Files” on page 28 - “DESIGNWARE_HOME Directory Structure” on page 29

Installation and Setup

This guide provides information on how to install and set up the DesignWare Cores SuperSpeed USB 3.0 Controller (also referred to as DWC_usb3). After you complete the installation procedure, you will work primarily with the Synopsys coreConsultant tool to configure and synthesize the controller, and simulate it in the provided verification environment (a Unit Test Environment [UTE] that allows you to run packaged tests on the controller).

**Attention**

If your DWC USB 3.0 Controller is already installed and you are performing only the setup procedures, check the IP Directory Component Detail page to confirm that you have the latest version.

Device: https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_device

Host: http://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_host

Dual Role Device: http://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_drd

Hub: http://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_hub

For Automotive Package:

Device: https://www.synopsys.com/dw/ipdir.php?c=dwc_ap_usb_3_0_device

Host: https://www.synopsys.com/dw/ipdir.php?c=dwc_ap_usb_3_0_host

Dual Role Device: https://www.synopsys.com/dw/ipdir.php?c=dwc_ap_usb_3_0_drd

1.1 Product Codes

Table 1-1 lists the products and product codes for the DWC_usb3 controller.

Table 1-1 Product Codes for DWC SuperSpeed USB 3.0 Controller

Component Name or Add-On	Product Code
Base Products	
DWC USB 3.0 Device-AHB ^a	6793-0
DWC USB 3.0 Host-AHB ^b	6897-0
DWC USB 3.0 Dual Role Device-AHB ^c	7567-0
DWC USB 3.0 Hub	6921-0
DWC AP USB 3.0 Device	C234-0
DWC AP USB 3.0 Host	C233-0
DWC AP USB 3.0 Dual Role Device	C232-0
Add-Ons	
DWC USB 3.0 Device-Isoch Add-On	7571-0
DWC USB 3.0 Host-Isoch Add-On	7593-0
DWC USB 3.0 DRD-Isoch Add-On	7568-0
DWC USB 3.0 Hibernation (LCA) Add-On	9228-0
DWC USB 3.0 HSIC Add-On	7387-0

a. DWC USB 3.0 Device-AHB includes DWC USB 3.0 Device-AXI Add-On and DWC USB 3.0 Device-Isoch Add-On.

b. DWC USB 3.0 Host-AHB includes DWC USB 3.0 Host-AXI Add-On and DWC USB 3.0 Host-Isoch Add-On.

c. DWC USB 3.0 Dual Role Device-AHB includes DWC USB 3.0 DRD-AXI Add-On and DWC USB 3.0 DRD-Isoch Add-On.

1.2 Accessing Product Documentation

Before you install the controller, you can download the full document set, including this document, the Databook, and Release Notes, at:

- Device
https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_device
- Host
https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_host
- Dual Role Device
https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_drd
- Hub
https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_hub

For Automotive Package:

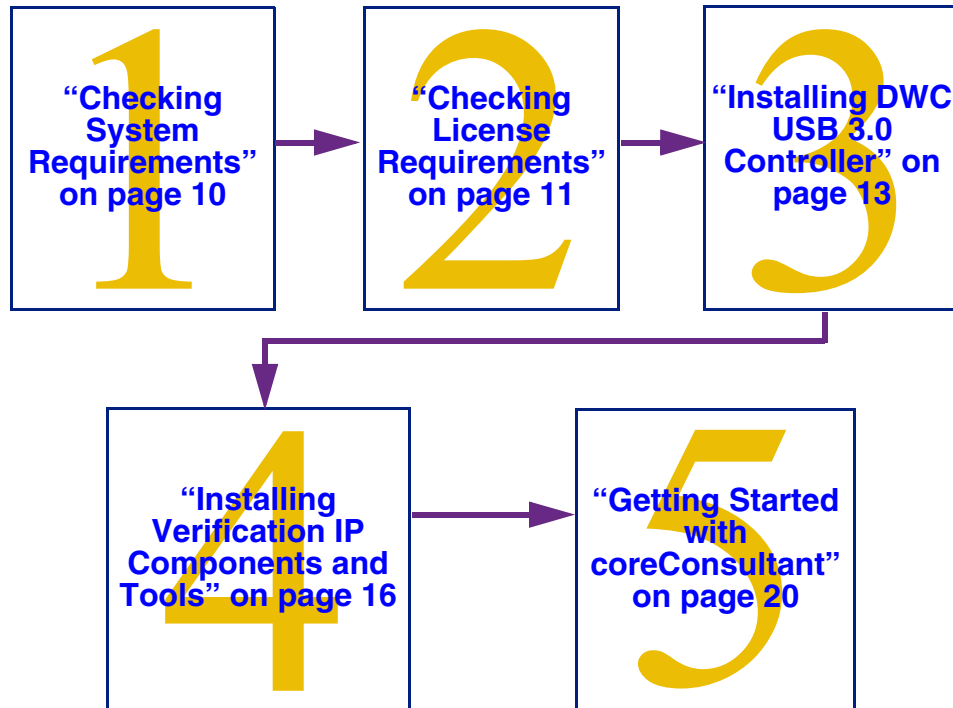
https://www.synopsys.com/dw/ipdir.php?c=dwc_ap_usb_3_0_device
https://www.synopsys.com/dw/ipdir.php?c=dwc_ap_usb_3_0_host
https://www.synopsys.com/dw/ipdir.php?c=dwc_ap_usb_3_0_drd

(You need to use a SolvNet ID and a valid license to download these documents.)

To access product documentation from your installation directory, you must first create a workspace using coreConsultant. Instructions for creating a workspace are provided in “[Getting Started with coreConsultant](#)” on page 20.

1.3 Process Overview

To use the DWC USB 3.0 Controller, you must follow the steps in this installation guide.



1.4 Checking System Requirements

Table 1-2 describes the system requirements for the DWC USB 3.0 Controller.

Table 1-2 System Requirements

Element	Requirement
Operating System	Linux: Red Hat Enterprise Linux 5.0/64 or later with all required patches installed Current information about supported operating system configurations and required patches can be found at the following web site: https://www.synopsys.com/support/licensing-installation-computeplatforms/licensing/scl-supported-os.html
Disk Space	The following numbers are the minimum disk space requirements for typical usage: ■ 300 MB available hard disk space for coreConsultant installation ■ 400 MB available hard disk space for DWC_usb3 installation ■ 14 GB available hard disk space for DWC_usb3 after simulation
Memory	■ 4 GB available swap space ■ 4 GB RAM



Note

You must have access to www.designware.com to download the release image. You can sign up to receive updates for any Designware® component through this web site as well.

1.5 Checking License Requirements

This section provides the required licenses information needed to use the DWC USB 3.0 Controller.

1.5.1 DWC USB 3.0 Controller Licenses

Synopsys provides the following licenses with the DWC USB 3.0 Controller:

- One DWC-USB30 license for running the product installation script (.run);
- Two AMBA licenses (supports two simultaneous simulations using AMBA models):
 - VIP-AMBA-AXI-SVT
 - OR
 - VIP-AMBA-AHB-SVT
- Two VIP-USB3-SVT licenses (supports two simultaneous simulations using USB3 VIP models);
- One RTL source code license for each USB 3.0 Controller base product and one add-on license for each add-on you have purchased, as shown in [Table 1-3](#).

Table 1-3 Licenses for DWC USB 3.0 Controller

DWC_usb3 Configuration	License Required
Base Products: ■ Device Only ■ Host Only ■ Dual Role Device ■ Hub	■ DWC-USB30-DEV-SRC, DWC-USB30-AXI, DWC-USB30-ISOC ■ DWC-USB30-HST-SRC, DWC-USB30-AXI, and DWC-USB30-ISOC ■ DWC-USB30-DEV-SRC, DWC-USB30-HST-SRC, DWC-USB30-AXI, and DWC-USB30-ISOC ■ DWC-USB30-HUB-SRC
Automotive Package ■ Device Only ■ Host Only ■ Dual Role Device	■ DWC-USB30-DEV-SRC, DWC-USB30-AP, DWC-USB30-AXI, DWC-USB30-ISOC ■ DWC-USB30-HST-SRC, DWC-USB30-AP, DWC-USB30-AXI, DWC-USB30-ISOC ■ DWC-USB30-DEV-SRC, DWC-USB30-HST-SRC, DWC-USB30-AP, DWC-USB30-AXI, DWC-USB30-ISOC
Add-Ons: ■ Hibernation	■ DWC-USB30-2PWR

1.5.2 Related Products

The following products are available for the DWC_usb3 controller through separate part numbers and written agreements.

- Device drivers
- Demo boards

For more information, contact your Synopsys sales representative.

1.5.3 Setting License File Environment Variable

Make sure that the DWC_usb3 license keys are installed on your license server. For more instructions about installing license keys, refer to the following site:

<https://www.synopsys.com/support/licensing-installation-computeplatforms/licensing.html>

A SolvNet ID is required.

You must set the SNPSLMD_LICENSE_FILE or LM_LICENSE_FILE environment variable to include a pointer to a license server that contains your DWC_usb3 license keys.

**Note**

You must set this environment variable BEFORE you install the .run file; otherwise, you will receive errors while installing the .run file.

To set the license file environment variable:

1. Set the license variable to include your license file or point to your license server:

```
% setenv SNPSLMD_LICENSE_FILE ${SNPSLMD_LICENSE_FILE}:<my_license_file|port@host>
```

or

```
% setenv LM_LICENSE_FILE ${LM_LICENSE_FILE}:<my_license_file|port@host>
```

2. Verify the license setup:

```
% echo $LM_LICENSE_FILE
```

```
% lmstat -c $LM_LICENSE_FILE -f <LICENSE_NAME>
```

or

```
% echo $SNPSLMD_LICENSE_FILE
```

```
% lmstat -c $SNPSLMD_LICENSE_FILE -f <LICENSE_NAME>
```

**Note**

Use these commands to check the license setup for all of your component's required licenses as described in [Table 1-3](#) on page 11.

1.6 Installing DWC USB 3.0 Controller

This section provides required steps and information to download and install the DWC_usb3 image.

1.6.1 Downloading DWC_usb3

The latest version of the DWC_usb3 is available through the DesignWare Download Web site.

1. Go to the following URL (a SolvNet ID and a valid product license are required):
 - Device
https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_device
 - Host
https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_host
 - Dual Role Device
https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_drd
 - Hub
https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_hub
2. Click on the link in the **Download** field to obtain the dw_iip_DWC_usb3_3.30b.run file.



Note

Save the .run file to a directory outside of any existing DESIGNWARE_HOME tree. For details on DESIGNWARE_HOME tree, see “Setting Up Your Environment”.

1.6.2 Setting Up Your Environment

Before you install the DWC USB 3.0 Controller’s .run file, you must set the following environment variables.

1. Set the DESIGNWARE_HOME environment variable to your installation directory:


```
% setenv DESIGNWARE_HOME <path to DWC_usb3_install_base_dir>
```
2. Set either the SNPSLMD_LICENSE_FILE or the LM_LICENSE_FILE variable, if you have not already done so.


```
% setenv SNPSLMD_LICENSE_FILE ${SNPSLMD_LICENSE_FILE}:<my_license_file|port@host>
or
% setenv LM_LICENSE_FILE ${LM_LICENSE_FILE}:<my_license_file|port@host>
```
3. Include the following in your PATH environment variable:


```
$DESIGNWARE_HOME/bin
```

\$DESIGNWARE_HOME/bin is required in your PATH environment variable as it contains the dwh_update, dwh_install, and dw_vip_setup scripts, which provide important diagnostic information in coreConsultant about the version of your DWC component(s). For more information, refer to [Table A-1](#) on page 22.

[Table B-1](#) on page 26 summarizes the environment variables you may need to set when using the DWC USB 3.0 Controller. The example .setup file on [page 27](#) shows how to configure the commonly used environment variables for the DWC USB 3.0 Controller.

1.6.3 Installing DWC_usb3

**Attention**

Replace example directory paths and placeholders in <...> with the correct paths and terms for your installation.

The .run file you downloaded is a self-extracting image that installs itself and performs some set-up operations. For more information about command options, enter:

```
% ./dw_iip_DWC_usb3_3.30b.run --help
```

For a list of the .run file command options, see [Appendix D](#) on page 28.

1. Change permissions on the downloaded .run file:

```
% chmod u+x dw_iip_DWC_usb3_3.30b.run
```

If you want to view the README information before performing the installation:

```
% dw_iip_DWC_usb3_3.30b.run --readme
```

2. Execute the .run file:

```
% dw_iip_DWC_usb3_3.30b.run
```

3. When prompted, enter the Project ID that you specified at the time of purchase.

For tips on how you can debug problems with installation, refer to [“Troubleshooting and Support”](#) on page 21.

1.6.4 Deliverables

This section describes the contents of the deliverable .run file.

1.6.4.1 Configurable USB 3.0 Controller

The USB 3.0 Controller can be configured using the coreConsultant tool. The configuration is dependent on license(s) purchased: Device, Host, DRD, or Hub. The coreConsultant generates Verilog RTL for the configured controller.

1.6.4.2 RTL

The Verilog RTL source is provided for the USB 3.0 Controller.

1.6.4.3 PHY GTECH and Behavioral Models

The Synopsys USB 3.0 PHY GTECH model of the digital layer and the behavioral model of the Analog layer is also supplied in the .../sim/phy/ss_gtech_10 directory. The USB 3.0 PHY is instantiated in the USB3_SOC_wrapper of the PVE as an example. The versions of the PHY tested in this release are:

- HSP=5.4c
- SSP=11.1
- PCS=1.6
- SDB=3.2

1.6.4.4 Customer Verification Environment

For the unit-level verification using coreConsultant, SoC Integration Example Testbench is used. The testbench is used to test interfaces and to show some protocol examples. AHB, AXI, and USB 3.0 Verification IPs are provided in the testbench and can be ported into your SoC-level testbench. USB 3.0 supports VCS.

1.6.4.5 Packaging

The USB 3.0 Controller is packaged for use with the coreConsultant tool.

1.6.4.6 Licensing Features

Licensing features and tool requirements for the USB 3.0 Controller are provided in [“Checking License Requirements”](#) on page 11.

1.7 Installing Verification IP Components and Tools

This section includes information for the supported tools that allow you to configure, synthesize, and verify the DWC USB 3.0 Controller. Instructions for installing the USB 3.0 Verification IP and coreConsultant are also provided in this section. If you need to install a synthesis or simulation tool, refer to the product documentation for that specific product.

1.7.1 Verifying Supported Verification IP Components and Tool Versions

The tools listed in [Table 1-4](#) have been tested for use with the DWC USB 3.0 Controller. The list of qualified tools is complete at the time of this release. Releases subsequent to this one may also be qualified.



Note Make sure that you use 64-bit binary for all the tools.

Table 1-4 Supported VIP and Tool Versions

Tool	Supported Version(s)	Description
coreConsultant	2017.12-SP1-2	Configures the controller and runs simulation and synthesis tools.
Design Compiler (DC) (optional)	2016.12-SP4	Synthesis, DFT (includes DFT Compiler).
Supported Simulators		
VCS Simulator	2017.03-SP2	Simulates the controller using the verification environment with VCS simulator.
Discovery VIP	■ AMBA_SVT N-2017.12 ■ SVT N-2017.12 ■ USB_SVT N-2017.12 ■ MPHY_SVT N-2017.12	Testbench components that are included in the USB 3.0 Controller image. AMBA_SVT and USB_SVT needs to be downloaded and installed separately. For instructions, refer to “Downloading and Installing Verification IP Components” on page 17.
Formality (optional)	2016.12-SP4	Equivalence checking
TetraMAX ATPG (optional)	2015.06-SP4	ATPG generation
Spyglass	2017.03-SP2	Lint and CDC
PrimeTime (optional)	2017.06	Static Timing Analysis checker



Note While no VHDL simulators are supported, you can generate a VHDL GTECH netlist of the design for use in your own environment.

1.7.2 Setting Environment Variables for Verification IP and Tools

Table 1-5 describes the environment variables required for the supported synthesis and simulation tools.

Table 1-5 Environment Variables for DWC USB 3.0 Controller

Tool	Environment Variable	Description
	LM_LICENSE_FILE	Path to license file for license manager
	LD_LIBRARY_PATH	Path to OS and tool libraries
	SNPSLMD_LICENSE_FILE	Path to license file for license manager
	DESIGNWARE_HOME	Path to the DWC_usb3 installation base directory
Synthesis Tools		
Design Compiler, Formality, PrimeTime, Synplicity, Other third-party tools	SYNOPSYS	Path to Synopsys tools tree (Design Compiler and others)
	PATH	#SYNOPSYS/bin \$SYNOPSYS/<platform>/syn/bin Path to <tool> bin directory, if used
Simulation Tools		
VCS	VCS_HOME	Path to VCS installation directory. Refer to the VCS documentation
	VCS_CC	Path to gcc compiler
	PATH	Include the absolute path to \$VCS_HOME/bin

1.7.3 Downloading and Installing Verification IP Components

To perform simulation, you must download and install the Discovery Verification IP for USB 3.0 (usb_svt). To download and install AMBA VIP, see the steps in [“Download and Install AMBA SVT Verification IP Components”](#) on page 18. When you install the AMBA VIP, the VMT components are installed as well. To download and install AMBA SVT, follow the steps in [“Download and Install AMBA SVT Verification IP Components”](#) on page 18.

If you do not want to run simulation, you can skip the steps listed in this subsection.



Note

- Separate VIP licenses are not required for any VIP components. They are provided with the DWC USB 3.0 Controller license.
- If coreConsultant is open when you install the VIP, to run simulations with the newly installed VIP, you must exit coreConsultant and open it again.

Download and Install USB 3.0 Verification IP Components

1. Download USB 3.0 Verification IP from the SolvNet Download Center, at:
<https://solvnet.synopsys.com/DownloadCenter/dc/product.jsp>
2. Select “VC VIP Library” from the list of available products under “My Product Releases”.
3. Select version N-2017.12 and follow the download instructions.
4. Download the `vip_usb_svt_N-2017.12.run` file.
5. Change permissions on the downloaded .run file:

```
% chmod u+x vip_usb_svt_N-2017.12.run
```
6. The `DESIGNWARE_HOME` environment variable must be set to the same directory where the `DWC_usb3` Controller is installed.
7. Execute the .run file:

```
% vip_usb_svt_N-2017.12.run
```

Download and Install AMBA SVT Verification IP Components

1. Download AMBA VIP from the SolvNet Download Center, at:
<https://solvnet.synopsys.com/DownloadCenter/dc/product.jsp>
2. Select “VC VIP Library” from the list of available products under “My Product Releases”.
3. Select version N-2017.12 and follow the download instructions.
4. Download the `vip_amba_svt_N-2017.12.run` file.
5. Change permissions on the downloaded .run file:

```
% chmod u+x vip_amba_svt_N-2017.12.run
```
6. The `DESIGNWARE_HOME` environment variable must be set to the same directory where the `DWC_usb3` Controller is installed.
7. Execute the .run file:

```
% vip_amba_svt_N-2017.12.run
```

Download and Install MPHY SVT Verification IP Components

1. Download MPHY SVT from the SolvNet Download Center, at:
<https://solvnet.synopsys.com/DownloadCenter/dc/product.jsp>
2. Select “VC VIP Library” from the list of available products under “My Product Releases”.
3. Select version N-2017.12 and follow the download instructions.
4. Download the `vip_mphy_svt_N-2017.12.run` file.
5. Change permissions on the downloaded `.run` file:

```
% chmod u+x vip_mphy_svt_N-2017.12.run
```
6. The `DESIGNWARE_HOME` environment variable must be set to the same directory where the `DWC_usb3` Controller is installed.
7. Execute the `.run` file:

```
% vip_mphy_svt_N-2017.12.run
```

1.7.4 Downloading and Installing coreConsultant

Synopsys’ `coreConsultant` is a tool that you use to configure, verify, and synthesize the `DWC USB 3.0` Controller. If you do not have a supported version of `coreConsultant` installed on your system or network, you must download and install it. If a supported version of `coreConsultant` is already installed, skip to “[Setting Environment Variables for Verification IP and Tools](#)” on page 17.

1. Download `coreConsultant` from the SolvNet Download Center, at:
<https://solvnet.synopsys.com/DownloadCenter/dc/product.jsp>
2. Select `coreConsultant` version 2017.12-SP1-2 and download the following three files:
 - a common file (`coretools_v2017.12-SP1-2_common.spf`)
 - a platform-specific file (`coretools_v2017.12-SP1-2_linux64.spf`)
 - a readme file (`coretools_INSTALL_README.txt`)
3. Follow the instructions in the readme file (`coretools_INSTALL_README.txt`) to install `coreConsultant`.

**Note**

To install `coreConsultant`, you also need to download Synopsys Installer as mentioned in the `coretools_INSTALL_README.txt` file.

1.8 Getting Started with coreConsultant

Now that you have completed installation and set up your environment, the next steps are to configure, synthesize, simulate, and export your DWC USB 3.0 Controller into your design.

Read the [DesignWare Cores SuperSpeed USB 3.0 Controller Databook](#) and [DesignWare Cores SuperSpeed USB 3.0 Controller User Guide](#) to learn about the DWC USB 3.0 Controller and the configuration options. The documents are available in the doc directory of your workspace (<workspace>/doc/DWC_usb3_user.pdf, /DWC_usb3_databook.pdf). The documents are also available in the coreConsultant Help Menu.

1. Create a workspace for the DWC USB 3.0 Controller.

A workspace is a local copy of your DWC_usb3 controller installation in which you can configure, verify, and synthesize your own implementation of the DWC_usb3.

To create a workspace:

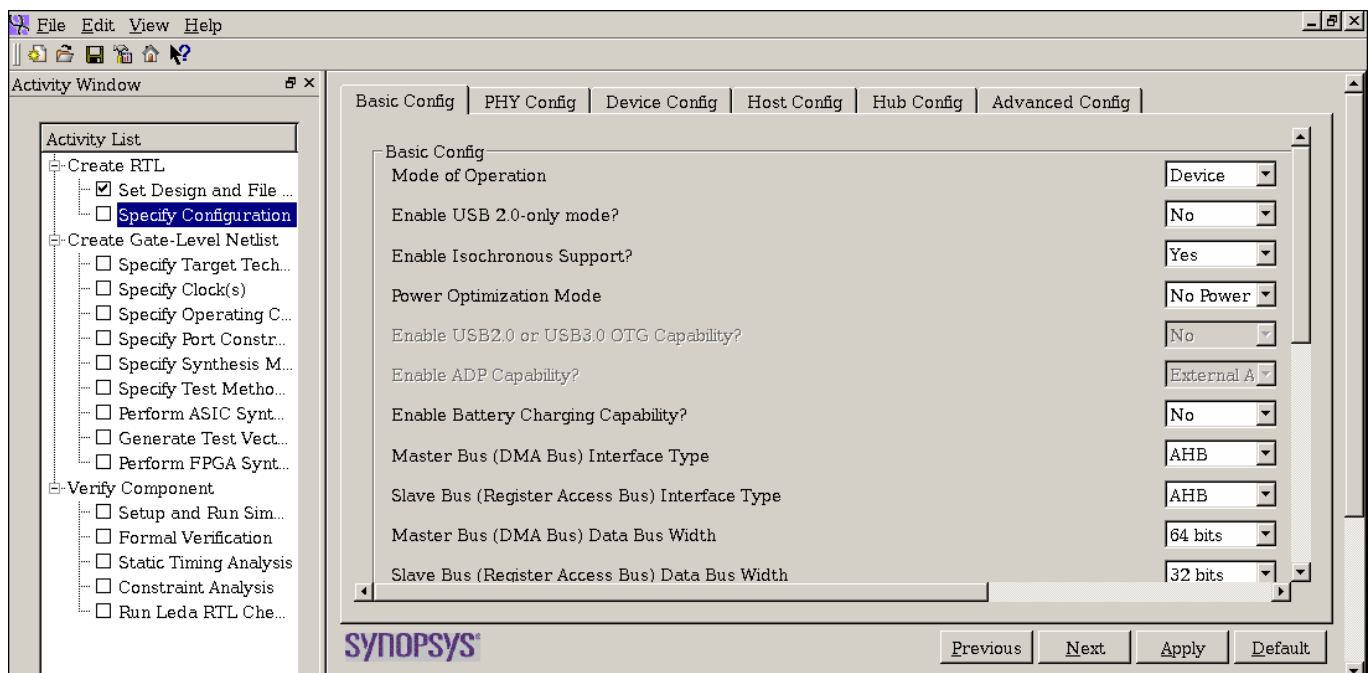
- a. Invoke coreConsultant:
% coreConsultant &
- b. Create a new workspace:
 - i. Select **File > New Workspace** in the coreConsultant console.
 - ii. Enter the requested information in the New Workspace dialog box.
 - iii. Optionally, for “Installed coreKit path”: select <install_dir>/iip/DWC_usb3/latest.
2. Complete the Specify Configuration activity using the default parameter values. Click Apply in the Specify Configuration dialog (Figure 1-1).

This creates the RTL for the default configuration and populates the product documentation (in <workspace>/doc).



Note You can create several workspaces so that you can experiment with different design alternatives.

Figure 1-1 Specify Configuration Activity



A

Troubleshooting and Support

This appendix provides troubleshooting information and details on how to contact customer support should you need further assistance during installation and set up.

A.1 Troubleshooting

This section provides troubleshooting tips if you encounter problems with licensing, installation, setting up your environment, and supported tools.

A.1.1 Installation

Q. *How can I make sure that the source installation was done properly?*

A. You can make sure that the source installation was done properly in one of three ways:

- Go to the \$DESIGNWARE_HOME/iip/component/latest/auxiliary directory and check to see that the .component file is present in this directory. For example, if you have a source license for the DWC USB 3.0 Controller, the .DWC_usb3 file should be present in the “auxiliary” directory.
- Configure the IP in coreConsultant and check the <workspace>/src directory for source RTL.
- The .run installer script writes out a log file that you can use to debug any issues with installation. The log file is located in the installation directory and is named as follows:

```
<image>_<yyyymmdd>_<hhmmss>_<processid>.log
```

By default, this log file is retained only if errors occur; it is removed if installation completes successfully. To retain the log file irrespective of the installation status, you can use the --keep-log switch with the .run installer script:

```
./dw_iip_DWC_usb3_3.30b.run --keep-log
```

**Note**

A source license gives you access to all of the files when you configure your controller.

Q: *How do I verify that I installed the DWC USB 3.0 Controller components correctly?*

A: Once you have correctly downloaded and installed the DWC USB 3.0 Controller product, the product deliverables are unpacked into the \$DESIGNWARE_HOME directory (environment variable you set that points to the DWC USB 3.0 Controller's installation root directory).

To view the contents of the DWC_usb3 directory in the \$DESIGNWARE_HOME path, enter the following command at a Unix prompt:

```
% ls -l $DESIGNWARE_HOME/iip/DWC_usb3/latest
```



Note

The latest directory is a symbolic link to the DWC USB 3.0 Controller/<current_version> directory.

Figure E-2 on page 30 illustrates the directory structure after you have installed the DWC USB 3.0 Controller product in the \$DESIGNWARE_HOME directory.

Q: *How can I verify the version of my installed DesignWare Cores product to make sure I have the latest version?*

A: To verify whether your DesignWare Cores products are current, use one of the following methods:

- Subscribe to MyDesignware notifications on a component basis, where you receive an email when a component updates or has new/updated STAR information.
- Use the dwh_update command (in \$DESIGNWARE_HOME/bin) to determine the versioned objects (components, tools) in your \$DESIGNWARE_HOME, and to manage updates to these entities.
- Enable automatic update checking in coreConsultant to check the components in your design against both your DesignWare Cores and the currently supported Synopsys components.

Table A-1 provides more information about these three features.

Table A-1 DesignWare Component Update Features

DesignWare Update Feature	Description
MyDesignWare Notifications	Enables you to receive product updates, technical articles, in-depth application notes and more for products of interest to you. You can add or remove selected subscriptions at any time. Sign-up through your SolvNet user account at: https://www.synopsys.com/dw/mydesignware.php myDesignWare subscriptions include : ■ DesignWare Technical Bulletin ■ DesignWare Component Notifications

Table A-1 DesignWare Component Update Features (Continued)

DesignWare Update Feature	Description
dwh_update Command	Located in \$DESIGNWARE_HOME/bin. Performs the following functions: ■ Downloads and installs updates to DesignWare images you may have installed previously ■ Provides a quick way to list the components and their versions in your \$DESIGNWARE_HOME tree To display usage instructions for the current version of dwh_update, execute the -help option: <pre>% dwh_update -help</pre> Synopsys' customer support representatives may ask you to run a dwh_update -info report on your \$DESIGNWARE_HOME tree to debug an installation problem. The following command string is a typical example invocation: <pre>% dwh_update -info 1 -r /remote/project_1/reports</pre>
coreTools Update Checking	When you complete the Specify Configuration activity in coreConsultant, it checks your component version(s) against the most recent version(s) available both for download from Synopsys, and in your local \$DESIGNWARE_HOME library. A report gives you newer version information, if available, and lists STARS created/fixed for the component(s) you are using. You can manually check at any time using Help > Check for IP Updates. For more information about Automatic/Manual IP update checks in coreConsultant, see "Component Update Checking" in the coreConsultant User Guide. NOTE: Components are not automatically updated; this operation only generates a report. You must make these component updates manually.

A.1.2 Tools

Q: I have a DesignWare license and installed the image properly. When I invoke coreConsultant, I get the following message. Why?

```
Command not found
```

A: You get this message because coreConsultant was not installed or the \$PATH is not updated with the path to the coreConsultant tool installation directory. It is not a part of an image and has to be installed separately. For more information about installing coreConsultant, refer to [“Downloading and Installing coreConsultant”](#) on page 19. The other possibility is not having DESIGNWARE_HOME/bin in the PATH.

Q: How can I tell if coreConsultant installed?

A: Issue the following command, which should return the path to your coreConsultant installation directory. If not, complete the steps in [“Downloading and Installing coreConsultant”](#) on page 19.

```
% which coreConsultant
```

Q: How can I test to see if my coreConsultant environment is set up correctly before I begin configuring my design?

A: You can run simulation and synthesis with the default configuration to make sure your environment is set up correctly to access your simulation and synthesis tools.

Q: How can I verify if my tools are correctly set up to work with coreConsultant?

A: After completing the DWC USB 3.0 Controller installation and setting the required environment variables, confirm system and coreConsultant access to the tools:

1. Check that you have access to the supported tools specified in [Table 1-4](#) on page 16, as follows:

```
% echo $SYNOPSYS
% which coreConsultant
% which <verilog_simulator> [vcs]
```

2. Set DESIGNWARE_HOME to the DWC USB 3.0 Controller installation base directory where you installed the DWC USB 3.0 Controller files, then invoke coreConsultant:

```
% setenv DESIGNWARE_HOME <DWC_usb3_install_base_dir>
% coreConsultant &
```

3. Click the DWC USB 3.0 Controller link to create a new configuration.
4. In the coreConsultant menu bar, select **Edit > Tool Installation Roots** to verify valid versions of all tools needed. These may include:
 - ☐ Design Compiler (dc_shell)
 - ☐ PrimeTime (pt_shell)
 - ☐ Formality (fm_shell)
 - ☐ TetraMax (tmax)
 - ☐ Synplify FPGA (synplify)



Note

Use 64-bit execution for all tools. To use the 64-bit execution, make sure that you select the 64-bit option in the Set tool Installation Roots dialog (**Edit -> Tool Installation Roots**).

5. Exit coreConsultant by selecting **File > Exit** from the menu bar.

A.2 Customer Support

To obtain support for your product, choose one of the following:

- First, prepare the following debug information, if applicable:
 - For environment setup problems or failures with configuration, simulation, or synthesis that occur within coreConsultant or coreAssembler, use the following menu entry:

File > Build Debug Tar-file

Check all the boxes in the dialog box that apply to your issue. This menu entry gathers all the Synopsys product data needed to begin debugging an issue and writes it to the file `<core tool startup directory>/debug.tar.gz`.
 - For simulation issues outside of coreConsultant or coreAssembler:
 - Create a waveforms file (such as VPD or VCD)
 - Identify the hierarchy path to the DesignWare instance
 - Identify the timestamp of any signals or locations in the waveforms that are not understood
- Then, contact the Support Center, with a description of your question and supply the above information, using one of the following methods:
 - *For fastest response*, use the SolvNet website. If you fill in your information as explained below, your issue is automatically routed to a support engineer who is experienced with your product. The **Sub Product** entry is critical for correct routing.

Go to http://solvnet.synopsys.com/support/open_case.action.

Provide the requested information, including:

- Product: DesignWare Cores
- Sub Product: USB 3.0 Device, USB 3.0 Host, USB 3.0 Hub, USB 3.0 DRD
- Version: 3.30b
- Problem Type:
- Priority:
- Title: Provide a short summary of the issue or list the error message you have encountered
- Description: For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood

After creating the case, attach any debug files you created in the previous step.

- Or, send an e-mail message to support_center@synopsys.com (your email will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
 - Include the Product name, Sub Product name, Product version, and Tool Version number in your e-mail (as identified above) so it can be routed correctly.
 - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood
 - Attach any debug files you created in the previous step.
- or, telephone your local support center:
 - North America:
Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
 - All other countries:
<https://www.synopsys.com/support/global-support-centers.html>

B

Environment Variables

This appendix provides a summary of the environment variables.

Table B-1 Environment Variables for DWC USB 3.0 Controller

Environment Variable	Description
DESIGNWARE_HOME	Path to the DWC USB 3.0 Controller installation base directory
SNPSLMD_LICENSE_FILE or LM_LICENSE_FILE	Path to license file for Synopsys tools such as Design Compiler, VCS, PrimeTime, any third-party tool)
SYNOPSYS	Path to Synopsys tools tree (Design Compiler, VCS, any third-party tool); not a required variable
LD_LIBRARY_PATH	Path to tools and OS libraries
VCS_HOME	Path to VCS installation directory
VCS_CC	Path to gcc compiler
VCS_BITMODE	Specifies 32-bit or 64-bit operation for VCS simulator
PATH	Must include the following paths: ■ \$DESIGNWARE_HOME/bin ■ \$SYNOPSYS/bin ■ \$SYNOPSYS/<platform>/syn/bin ■ Path to coreConsultant bin/ directory ■ Path to Formality bin/ directory (if used) ■ Paths to simulator binaries: \$VCS_HOME/bin

C

Example Setup File

The following example setup file demonstrates how to configure the required environment. The example shows all three supported simulation tools. Choose the commands for the tool you are using. In all cases, replace the example directory paths and placeholders (in <...>) with the correct paths and terms for your tool installations.

**Hint**

When you create your setup file, you might want to name it *.setup* and then place it in the root of your DWC_usb3 product installation directory.

Example C-1 Example of Setup File

```
# Synopsys Setup
setenv SYNOPSYS <synopsys_install_dir>
set path = ($path $SYNOPSYS/bin)
set path = ($path $SYNOPSYS/<platform>/syn/bin)

# DESIGNWARE_HOME
setenv DESIGNWARE_HOME <DWC_usb3_install_base_dir>

# coreConsultant Setup
set path = ($path <cC_install_dir>/bin)

# VCS Setup
setenv VCS_HOME <vcs_install_dir>
setenv VCS_CC /opt/SUNWspro/bin/cc
set path = ($path ${VCS_HOME}/bin)

# Formality Setup
set path = ($path <fm_install_dir>/bin )

# License Setup
setenv SNPSLMD_LICENSE_FILE ${SNPSLMD_LICENSE_FILE}:<my_license_file|port@host>
```

D

Options for .run Files

The `dw_iip_DWC_usb3_3.30b.run` file is a self-extracting image that contains the DWC USB 3.0 Controller product. Executing the `dw_iip_DWC_usb3_3.30b.run` file without optional switches extracts (installs) the image to a `$DESIGNWARE_HOME`, if available, or to a specified directory.

You can choose only one of the following optional switches:

- | | |
|---------------------------------|--|
| <code>--help</code> | Displays this message and exits. |
| <code>--readme</code> | Displays additional installation information, if available and exits. |
| <code>--check</code> | Checks image integrity and exits. |
| <code>--dir <path></code> | Installs product into <path> instead of <code>\$DESIGNWARE_HOME</code> |

If you are doing an installation, you can use either of the following options:

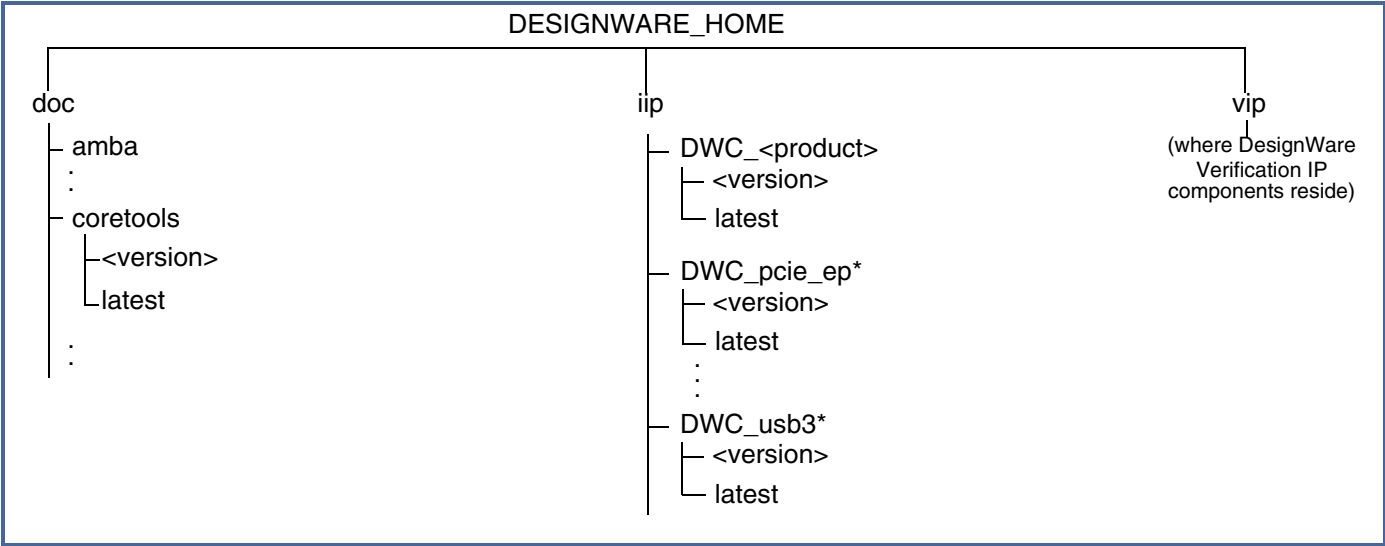
- | | |
|-------------------------|--|
| <code>--keep-log</code> | Keeps the installer log file regardless of the error count. By default, the log is deleted when there are no errors. |
| <code>--quiet</code> | Eliminates confirmation prompts. |

E

DESIGNWARE_HOME Directory Structure

This appendix provides two examples of a DESIGNWARE_HOME directory – a top-level view (Figure E-1) and the directory structure **after** you have installed the DWC USB 3.0 Controller into the DESIGNWARE_HOME directory (Figure E-2 on page 30). It also describes the directory structure of the DWC USB 3.0 Controller (Table E-1 on page 30).

Figure E-1 Example of a DESIGNWARE_HOME Directory



* Multiple DesignWare Cores, VIP, and DesignWare Library Synthesizable IP can reside in DESIGNWARE_HOME.

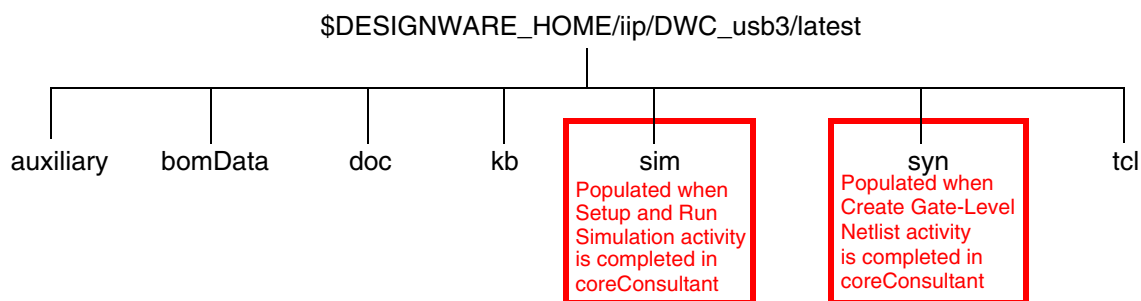
Figure E-2 Example of Directory Structure of \$DESIGNWARE_HOME/iip/DWC_usb3/latest

Table E-1 provides a description of the directories in \$DESIGNWARE_HOME/iip/DWC_usb3/latest directory.

Table E-1 \$DESIGNWARE_HOME/iip/DWC_usb3/latest

Directory	Description
auxiliary	Scripts and text files used by coreConsultant. Generated upon first creating a workspace in coreConsultant.
bomData	Build Of Material data used by coreConsultant to create the RTL based on the configuration selected.
doc	Contains the DWC USB 3.0 Controller product documentation, such as the databook and release notes (.pdf files)
kb	Contains knowledge base information used by coreConsultant. These are binary files containing information regarding the state of the design. Generated upon first creating a workspace; populated and updated throughout coreConsultant activities.
sim	Contains test stimulus and output files. Generated upon first creating a workspace; updated during Setup and Run Simulations activity in coreConsultant.
syn	Contains synthesis files for the DWC USB 3.0 Controller. Generated upon first creating a workspace; populated during the Synthesis activity and Formal Verification activity in coreConsultant.
tcl	Contains synthesis intent scripts. Generated upon first creating a workspace in coreConsultant.